# Osmotic Consequences (OC)

$$\dot{y} = +u_1 = c_1 x - C_2 f(y) T(t) y$$

$$\dot{x}: -u_1 + u_2 = -c_1 x + c_2 f(y) T(t) y + \frac{c_3 w}{x} - c_4 x$$

$$\frac{\partial \dot{y}}{\partial x} = c_1$$

$$\frac{\partial \dot{y}}{\partial y} = f'(y) T(t) y + f(y) T(t)$$

$$\boxed{c_1 - C_2 f'(y) T(t) y - f(y) T(t)}$$

$$\frac{\partial \dot{x}}{\partial x}: -c_1 - \frac{C_3 w}{x^2} - c_4$$

$$\frac{\partial \dot{x}}{\partial y} = C_2 f'(y) T(t) y - f(y) T(t)$$

$$\boxed{-\frac{C_3 w}{x^2} - C_2 f'(y) T(t) y - f(y) T(t) - c_1 - c_4}$$